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Chen et al.

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(54) **SEMICONDUCTOR PACKAGES AND METHODS OF MANUFACTURING THE SAME**

(58) **Field of Classification Search**
None
See application file for complete search history.

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(Continued)

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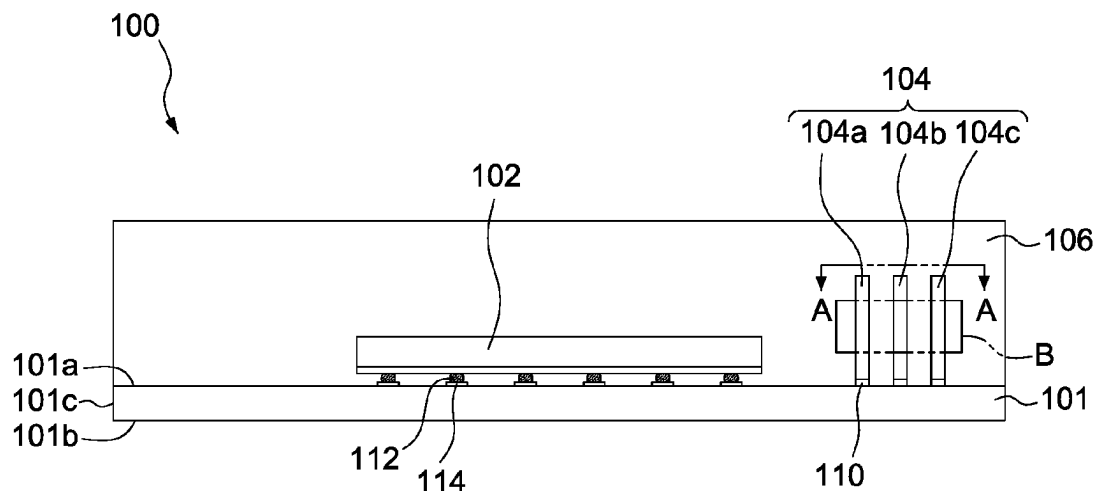
(51) **Int. Cl.**
H01L 23/552 (2006.01)
H01L 21/48 (2006.01)
(Continued)

(57) **ABSTRACT**

(52) **U.S. Cl.**
CPC **H01L 23/552** (2013.01); **H01L 21/4853** (2013.01); **H01L 21/565** (2013.01); **H01L 23/3121** (2013.01); **H01L 23/49838** (2013.01); **H01L 23/66** (2013.01); **H01L 24/16** (2013.01); **H01Q 1/2283** (2013.01); **H01Q 1/526** (2013.01);
(Continued)

A semiconductor package includes a substrate, a preformed feeding element, a preformed shielding element, and an encapsulant. The preformed feeding element is disposed on the substrate and the preformed feeding element is disposed on the substrate and adjacent to the preformed feeding element. The encapsulant encapsulates the preformed feeding element and the preformed shielding element.

9 Claims, 13 Drawing Sheets



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H01L 23/00 (2006.01)
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H01L 23/498 (2006.01)
H01L 23/66 (2006.01)
H01Q 1/22 (2006.01)
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- (52) **U.S. Cl.**
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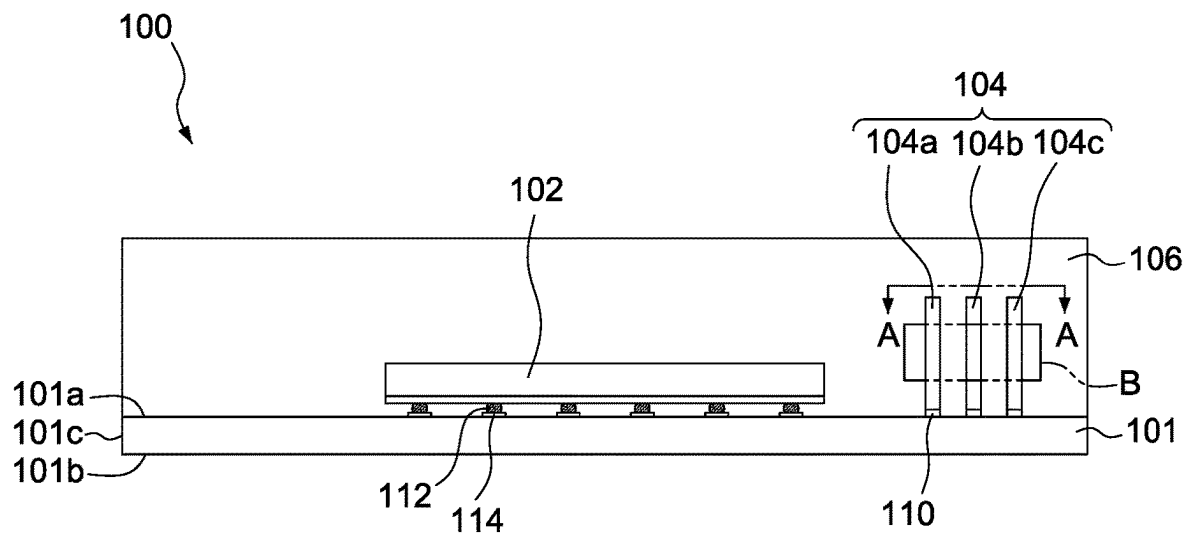


FIG. 1(a)

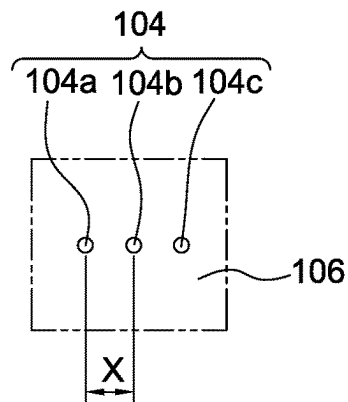


FIG. 1(b)

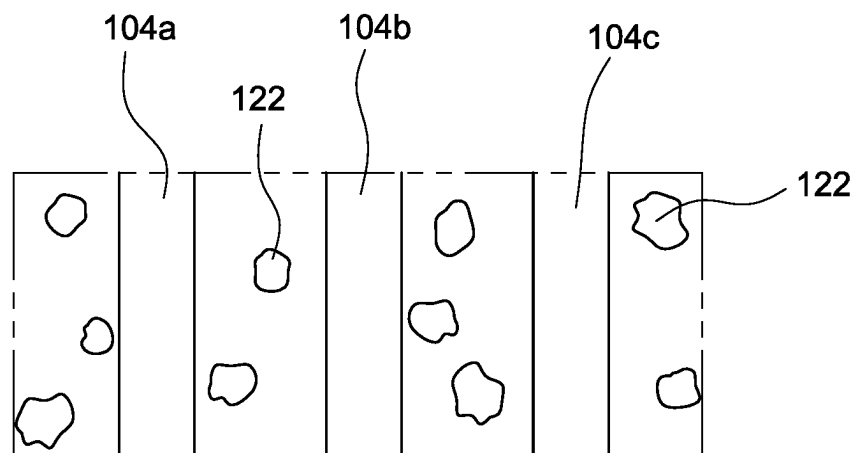


FIG. 1(c)

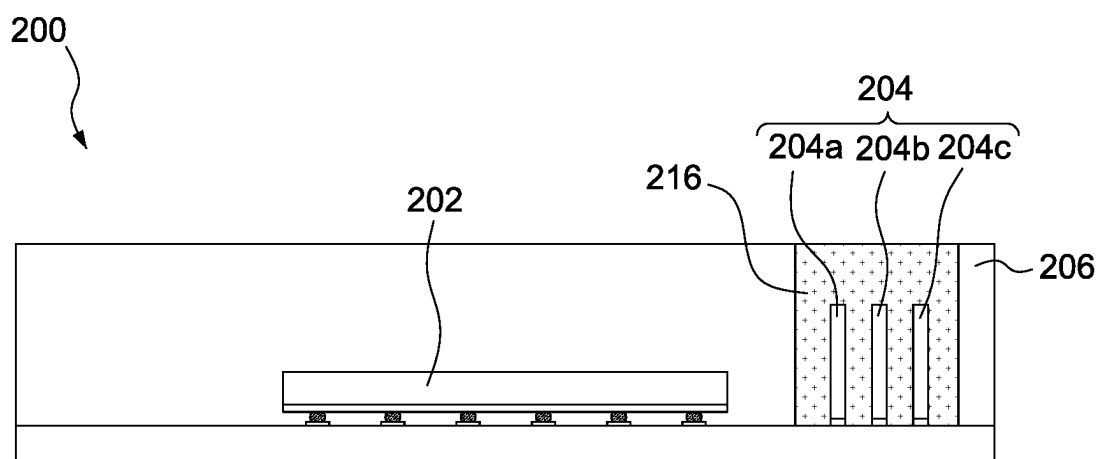


FIG. 2

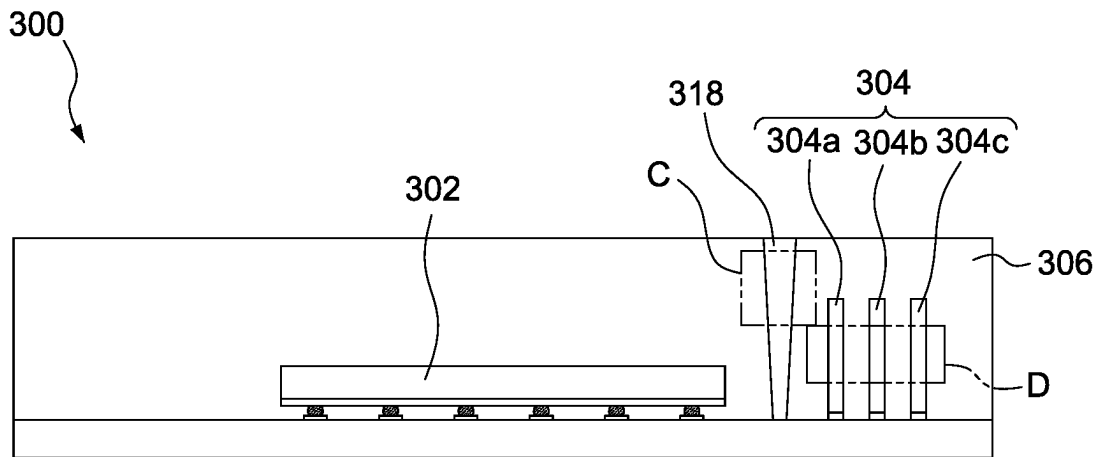


FIG. 3(a)

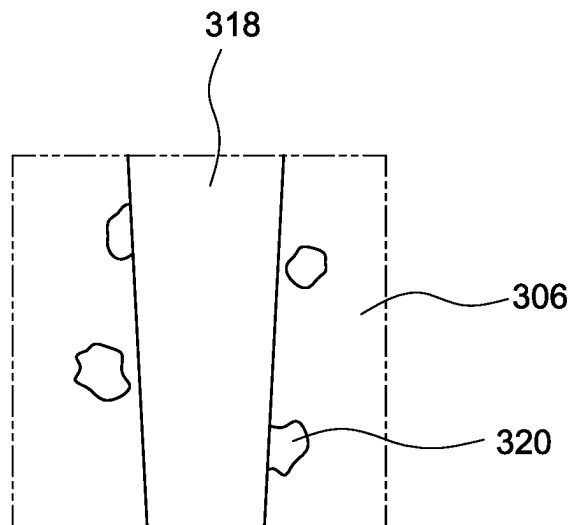


FIG. 3(b)

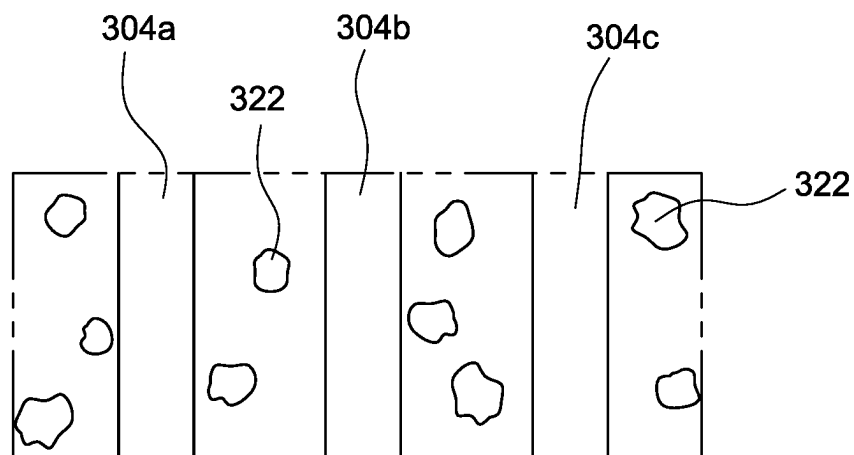


FIG. 3(c)

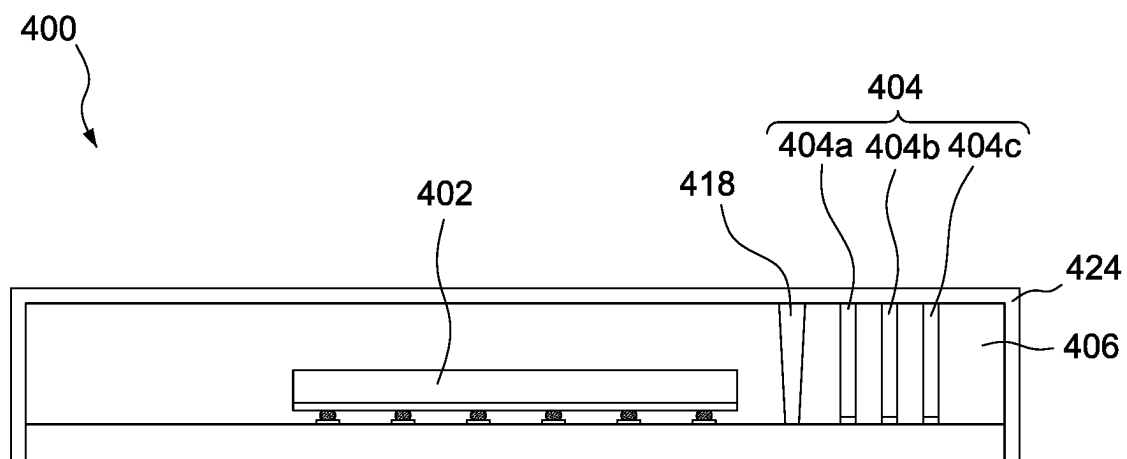


FIG. 4

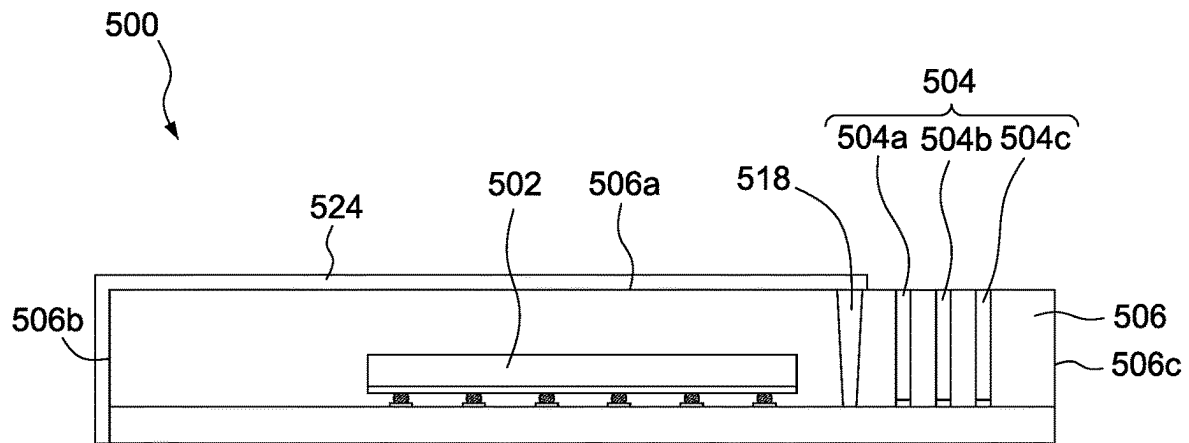


FIG. 5

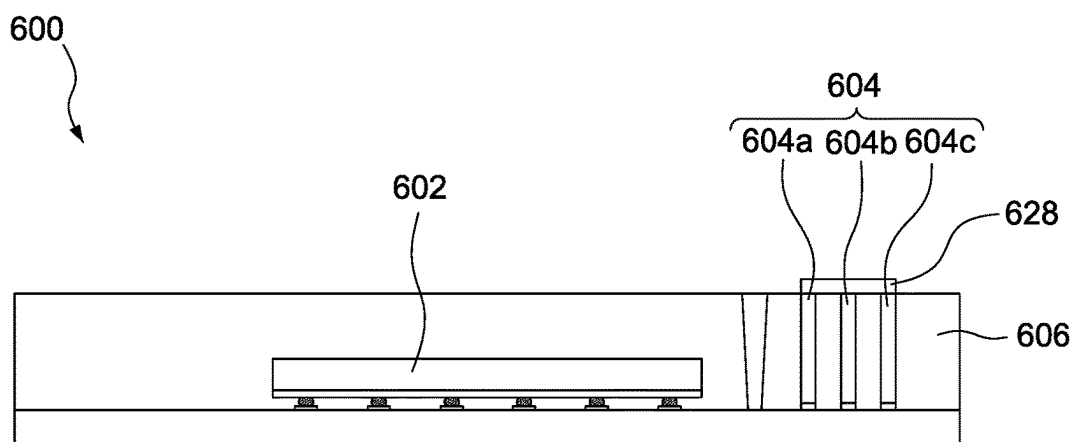


FIG. 6

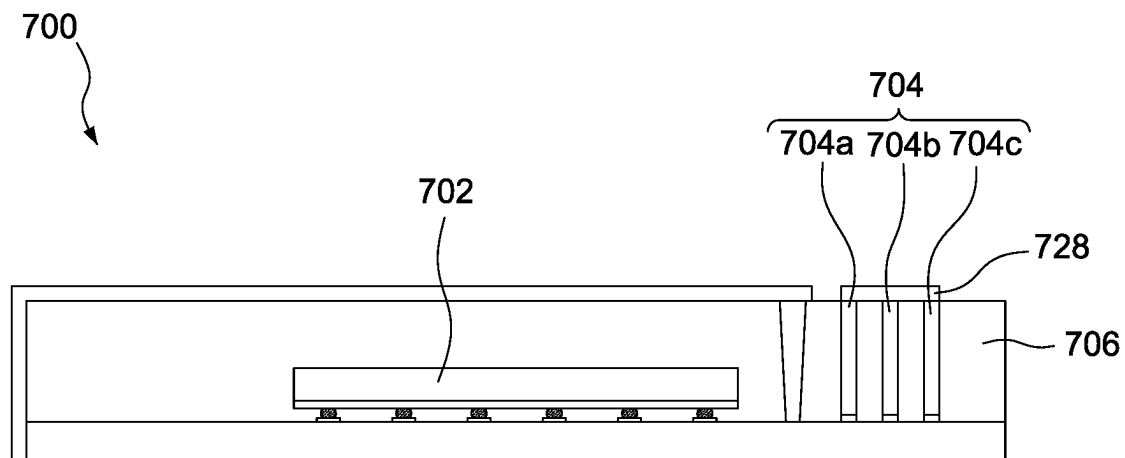


FIG. 7

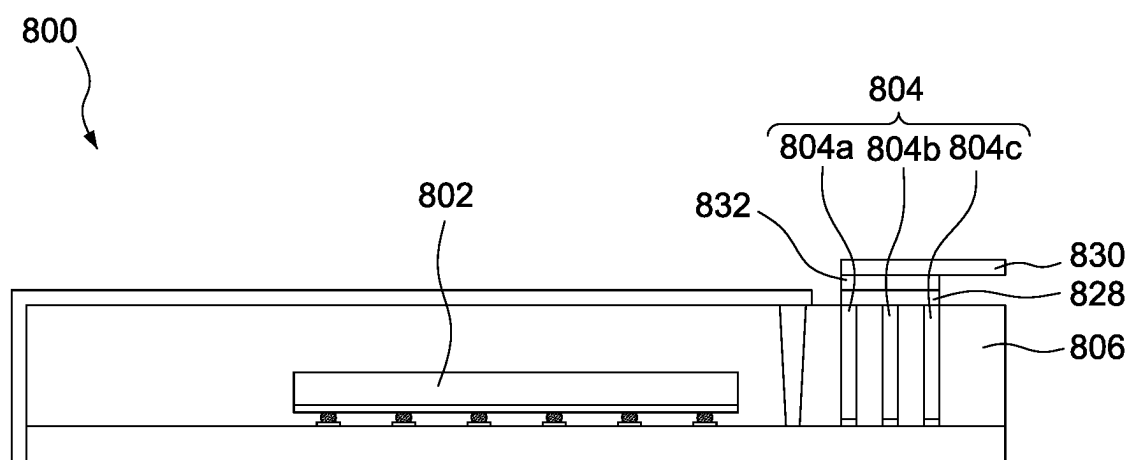


FIG. 8

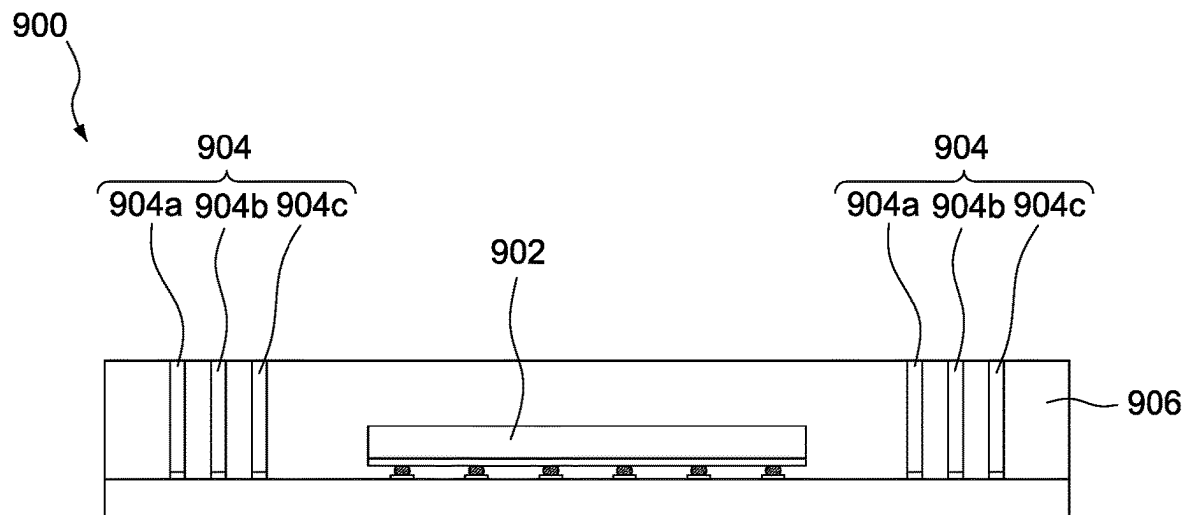


FIG. 9

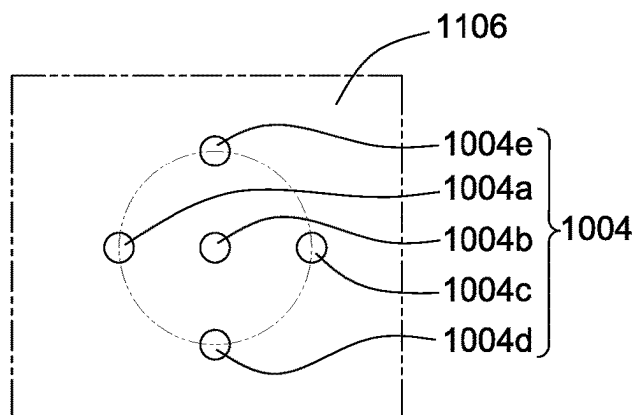


FIG. 10(a)

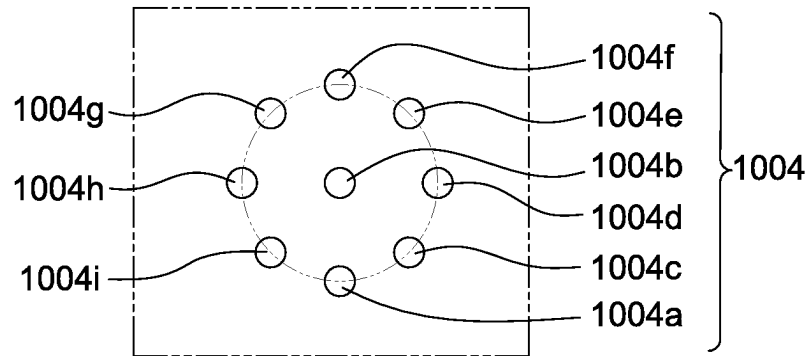


FIG. 10(b)

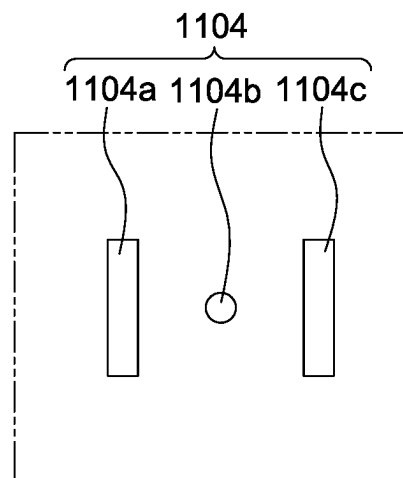


FIG. 11(a)

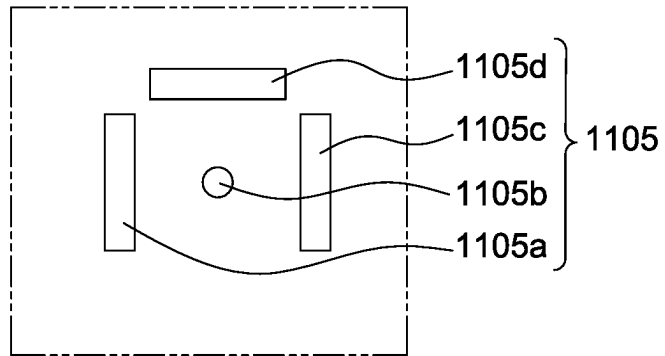


FIG. 11(b)

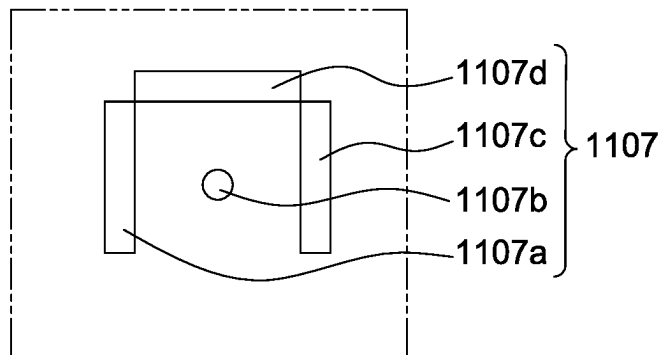


FIG. 11(c)

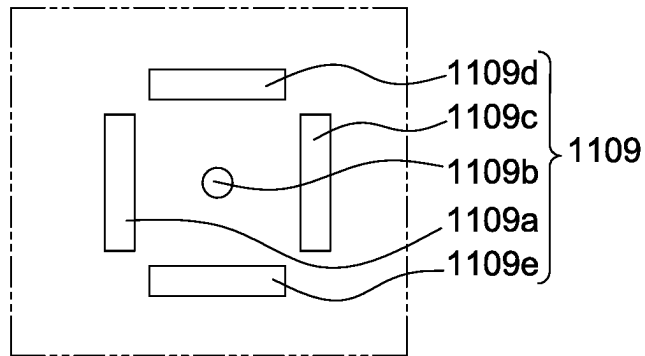


FIG. 11(d)

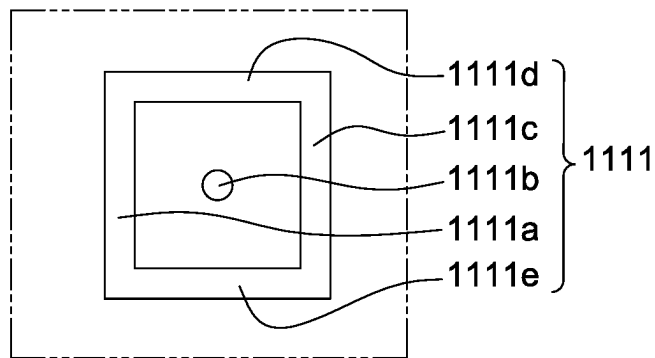


FIG. 11(e)

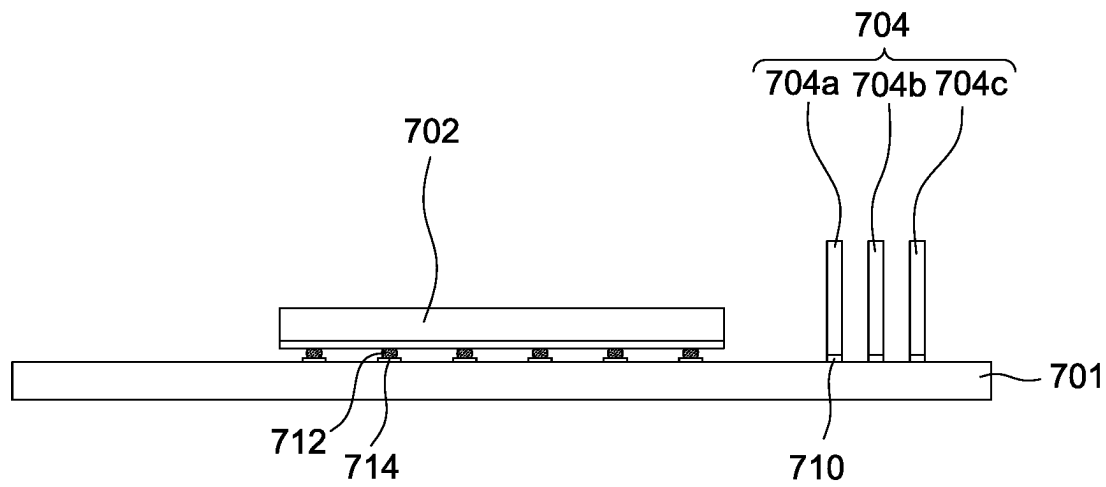


FIG. 12(a)

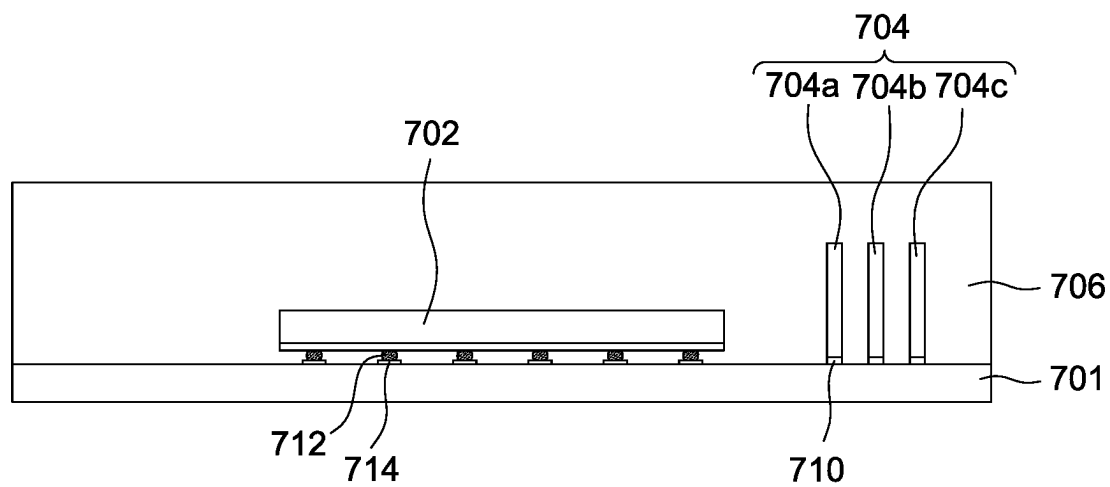


FIG. 12(b)

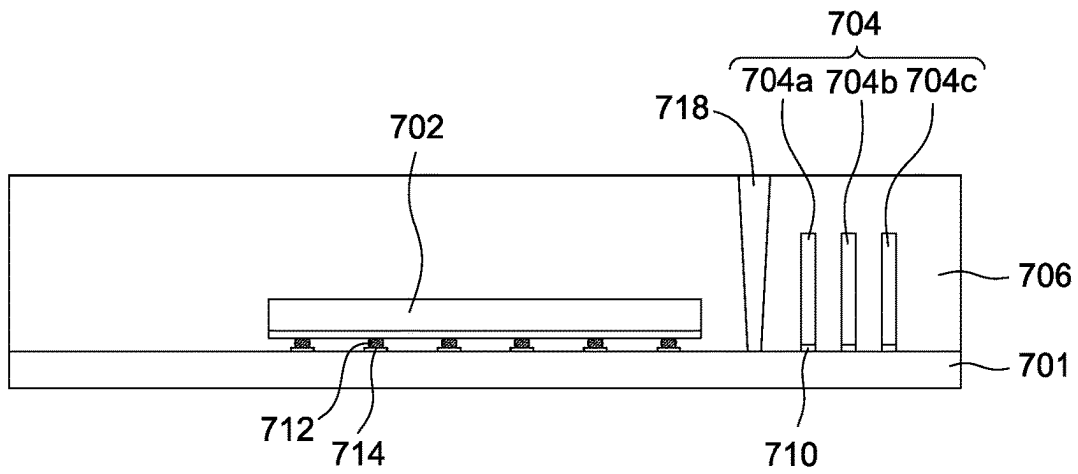


FIG. 12(c)

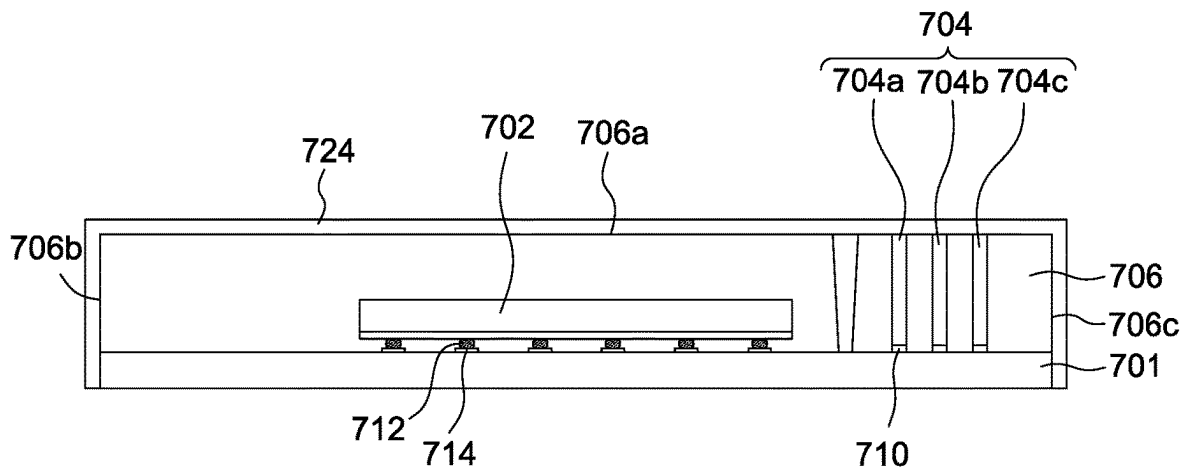


FIG. 12(d)

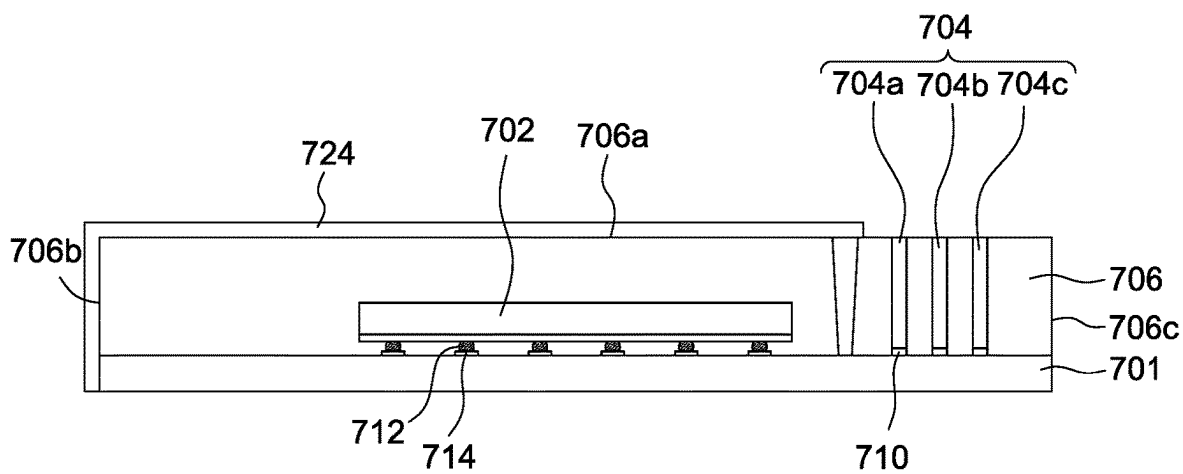


FIG. 12(e)

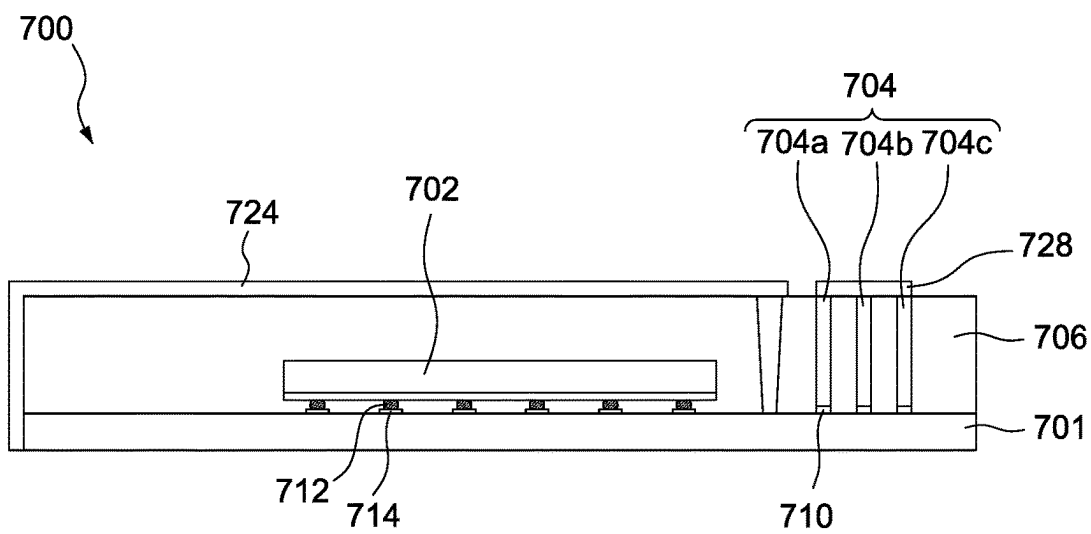


FIG. 12(f)

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SEMICONDUCTOR PACKAGES AND METHODS OF MANUFACTURING THE SAME

CROSS-REFERENCE TO RELATED APPLICATION

This application is a continuation of U.S. patent application Ser. No. 16/264,599 filed Jan. 31, 2019, now issued as U.S. Pat. No. 11,552,026, the contents of which is incorporated herein by reference in its entirety.

BACKGROUND

1. Technical Field

The present disclosure relates to semiconductor packages and method of manufacturing the same, and, more particularly, to semiconductor packages including at least two semiconductor components and method of manufacturing the same.

2. Description of the Related Art

There is a continuing desire to incorporate more than one semiconductor component into a single semiconductor package to reduce dimensions of the package. A semiconductor package incorporating multiple semiconductor components may be referred to as a system in package (SiP). Because semiconductor components in a semiconductor package specify electrical connections to an external environment, such electrical connections and the process for making the same are important in determining whether the semiconductor components can function properly or can achieve specified performances.

SUMMARY

In an embodiment, a semiconductor package includes a substrate; a preformed feeding element; a preformed shielding element; and an encapsulant. The preformed feeding element is disposed on the substrate and the preformed feeding element is disposed on the substrate and adjacent to the preformed feeding element. The encapsulant encapsulates the preformed feeding element and the preformed shielding element.

In an embodiment, a semiconductor package includes a substrate and a RF structure. The RF structure is disposed on the substrate and includes a feeding element and a shielding element adjacent to the feeding element, wherein a pitch from the feeding element to the shielding element is about $1000\text{ }\mu\text{m}$ to about $1500\text{ }\mu\text{m}$ with the insertion loss $\geq -0.5\text{ dB}$ under about 0.5 GHz to about 70 GHz .

In an embodiment, a semiconductor package includes a substrate and a RF structure. The RF structure is disposed on the substrate and includes a feeding element and a shielding element adjacent to the feeding element, wherein a pitch from the feeding element to the shielding element is about $0\text{ }\mu\text{m} < \text{Pitch} \leq \text{about } 800\text{ }\mu\text{m}$ with the return loss $\leq -10\text{ dB}$ under about 0.5 MHz to about 80 MHz .

In an embodiment, a semiconductor package includes a substrate and a RF structure. The RF structure is disposed on the substrate and includes a feeding element and a shielding element adjacent to the feeding element, wherein a pitch from the feeding element to the shielding element is about $0\text{ }\mu\text{m} < \text{Pitch} \leq \text{about } 800\text{ }\mu\text{m}$ with the insertion loss $2-0.5\text{ dB}$ under about 60 GHz to about 75 GHz .

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In an embodiment, a semiconductor package includes a substrate and a RF structure. The RF structure is disposed on the substrate and includes a feeding element and a shielding element adjacent to the feeding element, wherein a pitch from the feeding element to the shielding element is about $0\text{ }\mu\text{m} < \text{Pitch} \leq \text{about } 800\text{ }\mu\text{m}$ with the return loss $\leq -10\text{ dB}$ under about 0.5 MHz to about 80 MHz .

In an embodiment, a method of manufacturing includes (a) forming a RF structure on a substrate, the RF structure including a feeding element and a shielding element adjacent to the feeding element; and (b) molding the substrate, the feeding element and the shielding element.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1(a) illustrates a cross-sectional view of a semiconductor package according to an embodiment of the present disclosure.

FIG. 1(b) illustrates a top view of the second semiconductor component of FIG. 1(a) along with A-A line in accordance with an embodiment of the present disclosure.

FIG. 1(c) illustrates an enlarged view of an area B of the embodiment of a second semiconductor component illustrated in FIG. 1(a).

FIG. 2 illustrates a cross-sectional view of a semiconductor package according to an embodiment of the present disclosure.

FIG. 3(a) illustrates a cross-sectional view of a semiconductor package according to an embodiment of the present disclosure.

FIG. 3(b) illustrates an enlarged view of an area C of the conductive via of the embodiment of a third semiconductor component illustrated in FIG. 3(a).

FIG. 3(c) illustrates an enlarged view of an area D of the embodiment of a second semiconductor component illustrated in FIG. 3(a).

FIG. 4 illustrates a cross-sectional view of a semiconductor package according to an embodiment of the present disclosure.

FIG. 5 illustrates a cross-sectional view of a semiconductor package according to an embodiment of the present disclosure.

FIG. 6 illustrates a cross-sectional view of a semiconductor package according to an embodiment of the present disclosure.

FIG. 7 illustrates a cross-sectional view of a semiconductor package according to an embodiment of the present disclosure.

FIG. 8 illustrates a cross-sectional view of a semiconductor package according to an embodiment of the present disclosure.

FIG. 9 illustrates a cross-sectional view of a semiconductor package according to an embodiment of the present disclosure.

FIG. 10(a) illustrates a top view of a second semiconductor component in accordance with embodiments of the present disclosure.

FIG. 10(b) illustrates a top view of a second semiconductor component in accordance with embodiments of the present disclosure.

FIG. 11(a) illustrates a top view of a second semiconductor component in accordance with embodiments of the present disclosure.

FIG. 11(b) illustrates a top view of a second semiconductor component in accordance with embodiments of the present disclosure.

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FIG. 11(c) illustrates a top view of a second semiconductor component in accordance with embodiments of the present disclosure.

FIG. 11(d) illustrates a top view of a second semiconductor component in accordance with embodiments of the present disclosure.

FIG. 11(e) illustrates a top view of a second semiconductor component in accordance with embodiments of the present disclosure.

FIGs. 12(a), FIG. 12(b), FIG. 12(c), FIG. 12(d), FIG. 12(e), and FIG. 12(f) illustrate a method for manufacturing a semiconductor package in accordance with embodiments of the present disclosure.

DETAILED DESCRIPTION

Spatial descriptions, such as “top,” “side,” “over,” and so forth, are indicated with respect to the orientation shown in the figures unless otherwise specified. It should be understood that the spatial descriptions used herein are for purposes of illustration only, and that practical implementations of the structures described herein can be spatially arranged in any orientation or manner, provided that the merits of embodiments of this disclosure are not deviated by such arrangement.

FIG. 1(a) illustrates a cross-sectional view of a semiconductor package 100 according to an embodiment of the present disclosure. The semiconductor package 100 includes a substrate 101, a first semiconductor component 102, a second semiconductor component 104, and an encapsulant 106.

The substrate 101 has a first surface 101a, a second surface 101b, and a side surface 101c. The first surface 101a is opposite to the second surface 101b. The side surface 101c extends between the first surface 101a and the second surface 101b. In the embodiment illustrated in FIG. 1(a), the substrate 101 includes at least one bonding pad 114 disposed adjacent to the first surface 101a of the substrate 101. The bonding pad 114 may be, for example, a contact pad of a trace. In the embodiment of FIG. 1(a), the first surface 101a is an active surface, the bonding pad 114 is a contact pad, and the bonding pad 114 is disposed directly (e.g., in physical contact) on the first surface 101a of the substrate 101. The bonding pad 114 may include, for example, copper, gold, indium, tin, silver, palladium, osmium, iridium, ruthenium, titanium, magnesium, aluminum, cobalt, nickel, or zinc, other metals, metal alloys, or a combination of two or more thereof.

The first semiconductor component 102 is disposed on the first surface 101a of the substrate 101. The first semiconductor component 102 may be any semiconductor component including, for example, a chip, a package, an interposer, or a combination thereof. In the embodiment illustrated in FIG. 1(a), the first semiconductor component 102 is a chip including at least one conductive connector 112. The first conductive connector 112 contacts the bonding pad 114. The conductive connector 112 may be, for example, a pillar structure, which may include an under bump metallization (UBM) layer, a pillar, a barrier layer, a solder layer, or a combination of two or more thereof, or solder/stud bumps.

The second semiconductor component 104 is disposed on the first surface 101a of the substrate 101. The second semiconductor component 104 may be any semiconductor component including, for example, a RF structure. A RF structure may include at least one feeding element and at least one shielding element. According to the present disclosures, the feeding element and the shielding element were

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pre-formed so that their properties can be pre-determined and formed in any desired shape, for example, including square-like, triangular-like, round-like, rectangular-like, pentagonal-like, hexagonal-like, heptagonal-like, octagonal-like, trapezoidal-like, oval-like, rhombic-like, or parallelogram-like column shape. Unlike a feeding via or shielding via or ground via formed by photolithography in combination with etching (or drilling) and electroplating on a substrate, where the shape of the via fully depends on the drilling or etching technology and the properties of the via fully depends on the electroplating technology, the feeding element and the shielding element according to the embodiments of the present disclosures are pre-formed, for example, by molding, where their shape and properties can be controlled. Therefore, the voids in the feeding via or shielding via or ground via caused by electroplating can be avoided, which can reduce the signal loss.

In the embodiment illustrated in FIG. 1(a), the second semiconductor component 104 includes at least one pre-formed feeding element 104b disposed on the first surface 101a of the substrate 101 and at least one preformed shielding element 104a, 104c disposed on the first surface 101a of the substrate 101 and adjacent to the preformed feeding element 104b. The preformed feeding element 104b and the preformed shielding element 104a, 104c are spaced from each other by a distance. The shielding element 104a, 104c may include a plurality of pieces 104a, 104c spacing apart from each other. The feeding element 104b and the shielding element 104a, 104c may be disposed on the substrate 101 by a surface mount technology (SMT). In the embodiment illustrated in FIG. 1, the feeding element 104b and the shielding element 104a, 104c are disposed on the substrate 101 by a surface mount technology (SMT) with a solder paste 110.

The encapsulant 106 is disposed between the first semiconductor component 102 and the second semiconductor component 104. The encapsulant 106 encapsulates the pre-formed feeding element 104b and the preformed shielding element 104a, 104c. In the embodiment illustrated in FIG. 1(a), the preformed feeding element 104b defines a first space in the encapsulant 106 and the preformed shielding element 104a, 104c defines a second space in the encapsulant 106, where the encapsulant 106 surrounds them and accommodates the first space and the second space. The encapsulant 106 may extend from the preformed shielding element 104a to the first semiconductor component 102. The encapsulant 106 may cover the first semiconductor component 102, the preformed feeding element 104b, the preformed shielding element 104a, 104c, and the first surface 101a of the substrate 101, but not the side surface 101c of the substrate 101. The encapsulant 106 may be, for example, a solder mask (the material of which is, for example, polyimide (PI)), a passivation layer (the material of which is, for example, a metal oxide), or an underfill. The encapsulant 106 may include fillers, the material of which is, for example, silica and/or carbon for reducing stress on the die and warpage of a resulting semiconductor package.

FIG. 1(b) illustrates a top view of the second semiconductor component 104 of FIG. 1(a) along with A-A line in accordance with an embodiment of the present disclosure. The second semiconductor component 104 comprises one feeding element 104b and one shielding element 104a, 104c. The shielding element 104a, 104c surrounds the feeding element 104b. The shielding element 104a, 104c comprises two pieces 104a, 104c spacing apart from each other and includes at least one opening between them. The shielding element 104a, 104c are disposed adjacent to the feeding

element **104b** and are at the opposite sides of each other. The distance **X** from the center of the shielding element **104a**, **104b** to the center of the feeding element **104b** (a pitch) is determined according to the desired properties of the second semiconductor component **104**.

In an embodiment of the present disclosures, the pitch **X** can be about 1000 μm to about 1500 μm for a RF structure with an insertion loss ≥ -0.5 dB under about 0.5 GHz to about 70 GHz.

In an embodiment of the present disclosures, the pitch **X** can be about 1000 μm to about 1200 μm for a RF structure with an insertion loss ≥ -0.5 dB under about 0.5 GHz to about 70 GHz.

In an embodiment of the present disclosures, the pitch **X** can be about 1300 μm to about 1500 μm for a RF structure with an insertion loss ≥ -0.5 dB under about 0.5 GHz to about 70 GHz.

In an embodiment of the present disclosures, the pitch **X** can be about 1000 μm to about 1500 μm for a RF structure with a return loss ≤ -10 dB under about 0.5 MHz to about 80 MHz.

In an embodiment of the present disclosures, the pitch **X** can be about 1300 μm to 1500 μm with the return loss ≤ -10 dB under about 0.5 MHz to about 60 MHz.

In an embodiment of the present disclosures, the pitch **X** can be about 0 μm < Pitch ≤ 800 μm with the insertion loss ≥ -0.5 dB under about 60 GHz to about 75 GHz.

In an embodiment of the present disclosures, the pitch **X** can be about 0 μm < Pitch ≤ 800 μm with the return loss ≤ -10 dB under about 60 MHz to about 80 MHz.

The feeding element **104b** and the shielding element **104a**, **104c** are all surrounded by the encapsulant **106**. The feeding element **104b** and the shielding element **104a**, **104c** can be in different shape depending on the molding technology or the technology to be used to form them. In the embodiment illustrated in FIG. 1(b), an outer boundary of the feeding element **104b** is similar to an outer boundary of the shielding element **104a** and an outer boundary of the shielding element **104a** can be similar to an outer boundary of the opposite shielding element **104c**. In the embodiment illustrated in FIG. 1(b), the feeding element **104b** and the shielding elements **104a**, **104c** have a round-like column shape.

FIG. 1(c) illustrates an enlarged view of an area B of the embodiment of a second semiconductor component **104** illustrated in FIG. 1(a). In the embodiment illustrated in FIG. 1(c), the encapsulant **106** includes fillers **122**. The fillers **122** are adjacent to the feeding element **104b** and the shielding elements **104a**, **104c**. The fillers **122** may be in regular or irregular shape. In the embodiment illustrated in FIG. 1(c), the shape of the fillers **122** remain intact because the feeding element **104b** and the shielding elements **104a**, **104c** are preformed in accordance with the embodiments of the present disclosures rather than formed by photolithography in combination with etching (or drilling) and electroplating. Therefore, the fillers **122** in the encapsulant **106** are not damaged by the etching or drilling and the shape thereof can remain intact. Accordingly, the effect of the fillers **122**, such as reducing stress on the die and warpage of a resulting semiconductor package, will not be compromised and can be maintained.

FIG. 2 illustrates a cross-sectional view of a semiconductor package **200** according to an embodiment of the present disclosure. The semiconductor package **200** in FIG. 2 is similar to the semiconductor package **100** in FIG. 1, with differences including that the semiconductor package **200** includes a first encapsulant **216** and a second encapsulant

206. The first encapsulant **216** is adjacent to the second encapsulant **206**. The first encapsulant **216** encapsulates the second semiconductor component **204**. In particular, the first encapsulant **216** covers the preformed feeding element **204b**, the preformed shielding element **204a**, **204c**, and the first surface **201a** of the substrate **201**, but not the side surface **201c** of the substrate **201**. The second encapsulant **206** covers the first semiconductor component **202** and the first surface **201a** of the substrate **201**, but not the side surface **201c** of the substrate **201**. The first encapsulant **216** and the second encapsulant **206** are composed of different materials. The first encapsulant **216** and the second encapsulant **206** may include fillers, the material of which is, for example, silica and/or carbon for reducing stress on the die and warpage of a resulting semiconductor package.

FIG. 3(a) illustrates a cross-sectional view of a semiconductor package **300** according to an embodiment of the present disclosure. The semiconductor package **300** in FIG. 3 is similar to the semiconductor package **100** in FIG. 1, with differences including that the semiconductor package **300** further includes a third semiconductor component **318** disposed in the encapsulant **306**. The third semiconductor component **318** is disposed between the first semiconductor component **302** and the second semiconductor component **304**. The third semiconductor component **318** may be, for example, a compartment separating the first semiconductor component **302** from the second semiconductor component **304**, or a conductive via. In the embodiment illustrated in FIG. 3(a), the third semiconductor component **318** is a conductive via extending through the encapsulant **306** from the substrate **308** to the surface of the encapsulant **306** and formed by drilling (or etching) and electroplating.

FIG. 3(b) illustrates an enlarged view of an area C of the conductive via **318** of the embodiment of a third semiconductor component **318** illustrated in FIG. 3(a). FIG. 3(c) illustrates an enlarged view of an area D of the embodiment of a second semiconductor component **304** illustrated in FIG. 3(a). The encapsulant **306** includes fillers **320** adjacent to the conductive via **318** and fillers **322** adjacent to the feeding element **304b** and the shielding elements **304a**, **304c**. As described above for FIG. 1(c), the fillers **322** adjacent to the feeding element **304b** and the shielding elements **304a**, **304c** remain intact in shape and so are their effects. Contrary to those fillers **322** adjacent to the feeding element **304b** and the shielding elements **304a**, **304c**, fillers **320** adjacent to the conductive via **318** cannot remain intact in shape as they are damaged by a drilling or etching process for forming the conductive via. Accordingly, the effect of the fillers **320** adjacent to a via formed by a drilling or etching process, for example, in reducing stress or warpage of a resulting semiconductor package will be reduced. Therefore, comparing a semiconductor component which is preformed to that is formed by drilling (or etching) and electroplating, the efficacy of an encapsulant adjacent to them will be different and deteriorated.

FIG. 4 illustrates a cross-sectional view of a semiconductor package **400** according to an embodiment of the present disclosure. The semiconductor package **400** in FIG. 4 is similar to the semiconductor package **300** in FIG. 3(a), with differences including that the semiconductor package **400** includes a conductive layer **424** disposed on the encapsulant **406**. The conductive layer **424** covers the top surface **406a** of the encapsulant **406**, the side surfaces **406b**, **406c** of the encapsulant **406**, the exposed surface of the second semiconductor component **404** (including the feeding element **404b** and the shielding elements **404a**, **404c**), and the exposed surface of the third semiconductor component **418**.

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The conductive layer **424** may be, for example, a shielding layer or a conformal shielding layer.

FIG. **5** illustrates a cross-sectional view of a semiconductor package **500** according to an embodiment of the present disclosure. The semiconductor package **500** in FIG. **5** is similar to the semiconductor package **400** in FIG. **4**, with differences including that the conductive layer **524** disposed on the encapsulant **506** covers the top surface **506a** of the encapsulant **506**, a single one of the side surfaces **506b**, **506c** of the encapsulant **506**, the exposed surface of the third semiconductor component **518**, but not the exposed surface of the second semiconductor component **504** (including the feeding element **504b** and the shielding elements **504a**, **504c**).

FIG. **6** illustrates a cross-sectional view of a semiconductor package **600** according to an embodiment of the present disclosure. The semiconductor package **600** in FIG. **6** is similar to the semiconductor package **300** in FIG. **3(a)**, with differences including that the semiconductor package **600** includes a connector **628** disposed on the encapsulant **606**. The connector **628** is disposed adjacent to the second semiconductor component **604**. In the embodiment illustrated in FIG. **6**, the connector **628** is disposed on the feeding element **604b** and the shielding element **604a**, **604c**. The connector **628** may be, for example, a connector for connecting to an antenna layer.

FIG. **7** illustrates a cross-sectional view of a semiconductor package **700** according to an embodiment of the present disclosure. The semiconductor package **700** in FIG. **7** is similar to the semiconductor package **500** in FIG. **5**, with differences including that the semiconductor package **700** includes a connector **728** disposed on the encapsulant **706**. The connector **728** is disposed adjacent to the second semiconductor component **704**. In the embodiment illustrated in FIG. **7**, the connector **728** is disposed on the feeding element **704b** and the shielding element **704a**, **704c**. The connector **728** may be, for example, a connector for connecting to an antenna layer.

FIG. **8** illustrates a cross-sectional view of a semiconductor package **800** according to an embodiment of the present disclosure. The semiconductor package **800** in FIG. **8** is similar to the semiconductor package **700** in FIG. **7**, with differences including that the semiconductor package **800** includes an antenna layer **830** disposed adjacent to the feeding element **804b** and the shielding elements **804a**, **804c** and electrically connected to the feeding element **804b** and the shielding element **804a**, **804c**. In the embodiment illustrated in FIG. **8**, the antenna layer **830** includes a second connector **832** connecting to the first connector **828** disposed on the feeding element **804b** and the shielding elements **804a**, **804c**.

FIG. **9** illustrates a cross-sectional view of a semiconductor package **900** according to an embodiment of the present disclosure. The semiconductor package **900** in FIG. **9** is similar to the semiconductor package **100** in FIG. **1(a)**, with differences including that the semiconductor package **900** includes a third semiconductor component **934** disposed on the substrate **908** and the encapsulant **906** exposes the second semiconductor component **904** and the third semiconductor component **934**. The third semiconductor component **934** is disposed on the first surface **901a** of the substrate **901** and may be at the same side or opposite side of the second semiconductor component **904**. The third semiconductor component **934** may be any semiconductor component including, for example, a RF structure, a chip, a package, an interposer, or a combination thereof. The third semiconductor component **934** may be the same or different

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from the second semiconductor component **904**. In the embodiment illustrated in FIG. **9**, the third semiconductor component **934** is a RF structure including at least one preformed feeding element **934b** and at least one preformed shielding element **934a**, **934c** and disposed at the opposite side of the second semiconductor component **904**.

FIG. **10(a)** illustrates a top view of a second semiconductor component in accordance with embodiments of the present disclosure. The second semiconductor component **1004** comprises one feeding element **1004b** and four pieces of shielding element **1004a**, **1004c**, **1004d**, **1004e**. The four pieces of shielding element **1004a**, **1004c**, **1004d**, **1004e** are disposed adjacent to the feeding element **1004b**. The four pieces of shielding element **1004a**, **1004c**, **1004d**, **1004e** surround the feeding element **1004b**. The shielding element **1004a**, **1004c**, **1004d**, **1004e** includes an opening between each pieces **1004a**, **1004c**, **1004d**, **1004e**. The four pieces of shielding element **1004a**, **1004c**, **1004d**, **1004e** are spaced apart from each other by the encapsulant **1006**. The four pieces of shielding element **1004a**, **1004c**, **1004d**, **1004e** may surround the feeding element **1004b** in any arrangement, for example, in a square-like, round-like, rectangular-like, trapezoidal-like, oval-like, rhombic-like, or parallelogram-like arrangement. In the embodiment illustrated in FIG. **10(a)**, the four pieces of shielding element **1004a**, **1004c**, **1004d**, **1004e** surround the feeding element **1004b** in a round-like arrangement.

FIG. **10(b)** illustrates a top view of a second semiconductor component in accordance with embodiments of the present disclosure. The second semiconductor component **1005** in FIG. **10(b)** is similar to the second semiconductor component **1004** in FIG. **10(a)**, with differences including that the second semiconductor component **1005** includes eight pieces of shielding element **1004a**, **1004c**, **1004d**, **1004e**, **1004f**, **1004g**, **1004h**, **1004i** disposed adjacent to the feeding element **1004b**. The eight pieces of shielding element **1004a**, **1004c**, **1004d**, **1004e**, **1004f**, **1004g**, **1004h**, **1004i** surround the feeding element **1004b** in a round-like arrangement.

FIG. **11(a)** illustrates a top view of a second semiconductor component in accordance with embodiments of the present disclosure. The second semiconductor component **1104** in FIG. **11(a)** is similar to the second semiconductor component **104** in FIG. **1(b)**, with differences including that the shielding element **1104a**, **1104c** are preformed in a rectangular-like column shape.

FIG. **11(b)** illustrates a top view of a second semiconductor component in accordance with embodiments of the present disclosure. The second semiconductor component **1105** in FIG. **11(b)** is similar to the second semiconductor component **1104** in FIG. **11(a)**, with differences including that the second semiconductor component **1105** includes three pieces of shielding element **1105a**, **1105c**, **1105d** disposed adjacent to the feeding element **1105b**. The three pieces of shielding element **1105a**, **1105c**, **1105d** surround the feeding element **1105b**. The three pieces of shielding element **1105a**, **1105c**, **1105d** may surround about $\frac{3}{4}$ of the surroundings of the feeding element **1105b**.

FIG. **11(c)** illustrates a top view of a second semiconductor component in accordance with embodiments of the present disclosure. The second semiconductor component **1107** in FIG. **11(c)** is similar to the second semiconductor component **1105** in FIG. **11(b)**, with differences including that the three pieces of shielding element **1107a**, **1107c**, **1107d** contact each other and there is no opening between each of them.

FIG. 11(d) illustrates a top view of a second semiconductor component in accordance with embodiments of the present disclosure. The second semiconductor component 1109 in FIG. 11(c) is similar to the second semiconductor component 1105 in FIG. 11(b), with differences including that there are four pieces of shielding element 1109a, 1109c, 1109d, 1109e surround the feeding element 1109b. The four pieces of shielding element 1109a, 1109c, 1109d, 1109e surround the feeding element 1109b in a square-like arrangement.

FIG. 11(e) illustrates a top view of a second semiconductor component in accordance with embodiments of the present disclosure. The second semiconductor component 1111 in FIG. 11(e) is similar to the second semiconductor component 1109 in FIG. 11(d), with differences including that the four pieces of shielding element 1111a, 1111c, 1111d, 1111e contact each other and there is no opening between each of them.

FIGS. 12(a)-12(b) illustrate a method for manufacturing a semiconductor package such as the semiconductor package 100 of FIG. 1(a). FIGS. 12(a)-12(c) illustrate a method for manufacturing a semiconductor package such as the semiconductor package 300 of FIG. 3(a). FIGS. 12(a)-12(d) illustrate a method for manufacturing a semiconductor package such as the semiconductor package 400 of FIG. 4. FIGS. 12(a)-12(e) illustrate a method for manufacturing a semiconductor package such as the semiconductor package 500 of FIG. 5. FIGS. 12(a)-12(f) illustrate a method for manufacturing a semiconductor package such as the semiconductor package 700 of FIG. 7.

Referring to FIG. 12(a), a first semiconductor component 702 and a second semiconductor component 704 are provided on a substrate 701. The first semiconductor component 702 is a chip including at least one conductive connector 712. The second semiconductor component 704 is a RF structure including at least one feeding element 704b and at least one shielding element 704a, 704c. The substrate 701 includes at least one bonding pad 714. The shielding element comprises two pieces 704a, 704c. The feeding element 704b and the shielding element 704a, 704c were preformed in a round-like column shape by a molding technology before being provided on the substrate 701. The preformed feeding element 704b and the preformed shielding element 704c are disposed on the substrate 701 by a surface mounted technology (SMT). The temperature of the SMT process is preferably controlled below about 200° C. The working temperature of the solder paste 710 is preferably below about 200° C. in the SMT process.

Referring to FIG. 12(b), an encapsulant 706 is disposed between the first semiconductor component 702 and the second semiconductor component 704 by, for example, molding. The encapsulant 706 encapsulates the first semiconductor component 702, the preformed feeding element 704b and the preformed shielding element 704a, 704c. The encapsulant 706 covers the first semiconductor component 702, the preformed feeding element 704b and the preformed shielding element 704a, 704c, surrounds the preformed feeding element 704b and the preformed shielding element 704a, 704c, and extends from the preformed shielding element 704a to the first semiconductor component 702. The encapsulant 706 may be, for example, a solder mask (the material of which is, for example, polyimide (PI)) or a passivation layer (the material of which is, for example, a metal oxide) or an underfill. The encapsulant 706 may include a filler, the material of which is, for example, silica and/or carbon.

Referring to FIG. 12(c), a third semiconductor component 718 is disposed in the encapsulant 706. The third semiconductor component 718 is disposed between the first semiconductor component 702 and the second semiconductor component 704. The third semiconductor component 718 may be, for example, a compartment separating the first semiconductor component 702 from the second semiconductor component 704, or a conductive via. The encapsulant 706 defines the location for forming the compartment or the conductive via. The compartment or conductive via may be formed by drilling (or etching) and electroplating. Accordingly, as described above for the encapsulant 706 including fillers adjacent to the compartment or conductive via 718, they cannot remain intact in shape as they are damaged by the drilling or etching process and therefore their effects in the encapsulant 706 as fillers will be deteriorated.

Referring to FIG. 12(d), the encapsulant 706 is ground to expose the second semiconductor component 704. Subsequently, a separation technique (e.g., sawing) is performed to obtain individual semiconductor packages such as the semiconductor package 300 of FIG. 3(a). A conductive layer 724 is disposed on the encapsulant 706 by, for example, an electroplating technology where it covers the top surface 706a of the encapsulant 706, the side surfaces 706b, 706c of the encapsulant 706, the exposed surface of the second semiconductor component 704 (including the feeding element 704b and the shielding elements 704a, 704c), and the exposed surface of the third semiconductor component 718. The conductive layer 724 may be, for example, a shielding layer or a conformal shielding layer.

Referring to FIG. 12(e), the conductive layer 724 is disposed on the encapsulant 706 where it selectively covers the top surface 706a of the encapsulant 706, one of the side surfaces 706b, 706c of the encapsulant 706, the exposed surface of the third semiconductor component 718 except for the exposed surface of the second semiconductor component 704. The conductive layer 724 may be, for example, a shielding layer or a conformal shielding layer, and be formed by photolithography in combination with etching and electroplating or physical vapor deposition.

Referring to FIG. 12(f), a connector 728 is disposed on the encapsulant 706. The connector 728 is disposed adjacent to the second semiconductor component 704. In the embodiment illustrated in FIG. 7, the connector 728 is disposed on the feeding element 704b and the shielding elements 704a, 704c. The connector 728 may be, for example, a connector for connecting to an antenna layer. The connector 728 may be formed by photolithography in combination with etching and electroplating or physical vapor deposition.

As used herein and not otherwise defined, the term “about” are used to describe and account for small variations. When used in conjunction with an event or circumstance, the term can encompass instances in which the event or circumstance occurs precisely as well as instances in which the event or circumstance occurs to a close approximation. For example, when used in conjunction with a numerical value, the term can encompass a range of variation of less than or equal to +10% of that numerical value, such as less than or equal to +5%, less than or equal to +4%, less than or equal to +3%, less than or equal to +2%, less than or equal to +1%, less than or equal to +0.5%, less than or equal to +0.1%, or less than or equal to +0.05%.

While the present disclosure has been described and illustrated with reference to specific embodiments thereof, these descriptions and illustrations are not limiting. It should be understood by those skilled in the art that various changes may be made and equivalents may be substituted without

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departing from the true spirit and scope of the present disclosure as defined by the appended claims. The illustrations may not necessarily be drawn to scale. There may be distinctions between the artistic renditions in the present disclosure and the actual apparatus due to manufacturing processes and tolerances. There may be other embodiments of the present disclosure which are not specifically illustrated. The specification and the drawings are to be regarded as illustrative rather than restrictive. Modifications may be made to adapt a particular situation, material, composition of matter, method, or process to the objective, spirit and scope of the present disclosure. All such modifications are intended to be within the scope of the claims appended hereto. While the methods disclosed herein have been described with reference to particular operations performed in a particular order, it will be understood that these operations may be combined, sub-divided, or re-ordered to form an equivalent method without departing from the teachings of the present disclosure. Accordingly, unless specifically indicated herein, the order and grouping of the operations are not limitations.

What is claimed is:

1. A semiconductor package, comprising:
a substrate;
an electronic component disposed over the substrate;
a radio frequency (RF) structure disposed over the substrate;
a first encapsulant disposed over the substrate and encapsulating the RF structure; and
a second encapsulant disposed over the substrate and in contact with the first encapsulant, wherein an interface is between the first encapsulant and the second encapsulant, wherein a top surface of the first encapsulant is parallel to a top surface of the second encapsulant, and wherein the RF structure comprises a feeding element and a plurality of shielding elements surrounding the feeding element.
2. The semiconductor package of claim 1, wherein a lateral surface of the RF structure is completely covered by the first encapsulant.
3. The semiconductor package of claim 1, wherein a first distance between a first shielding element of the plurality of shielding elements and a first lateral surface of the first encapsulant is less than a second distance between the feeding element and a second lateral surface of the first encapsulant opposite to the first lateral surface.
4. The semiconductor package of claim 1, wherein the plurality of shielding elements are in a round-like arrangement in a top view.

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5. The semiconductor package of claim 1, wherein the plurality of shielding elements are in a square-like arrangement in a top view.

6. A semiconductor package, comprising:

- a substrate;
- an electronic component disposed over the substrate;
- a feeding element disposed over the substrate;
- a first shielding element disposed over the substrate and adjacent to the feeding element;
- an encapsulant encapsulating the feeding element and exposing a top surface of the feeding element; and
- a shielding layer over the encapsulant, wherein a projection of the first shielding element non-overlaps the shielding layer vertically,

wherein the first shielding element and the feeding element are between an edge of the substrate and the electronic component, wherein the first shielding element is closer to the edge of the substrate than the feeding element is.

7. The semiconductor package of claim 6, wherein the first shielding element is a hollow square, and wherein the feeding element is located within the first shielding element in a top view.

8. A semiconductor package, comprising:

- a substrate;
- an electronic component disposed over the substrate;
- a feeding element disposed over the substrate;
- a first shielding element disposed over the substrate and adjacent to the feeding element;
- an encapsulant encapsulating the feeding element and exposing a top surface of the feeding element;
- a shielding layer over the encapsulant, wherein a projection of the first shielding element non-overlaps the shielding layer vertically;
- a second shielding element adjacent to the feeding element and opposite to the first shielding element with respect to the feeding element; and
- a third shielding element adjacent to the feeding element and between the first shielding element and the second shielding element.

9. The semiconductor package of claim 8, wherein a first imaginary line defined by the feeding element and the first shielding element is perpendicular to a second imaginary line defined by the feeding element and the third shielding element in a top view.

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